# Unsupervised Anomaly Detection in Credit Card Transactions

**Course:** INT396 - Unsupervised Learning (L:2 T:0 P:2 Credits:3)
**Session:** 2025-26

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## 📌 Project Overview

This project aims to detect **fraudulent credit card transactions** using **Unsupervised Machine Learning** techniques. Since fraudulent cases are extremely rare (~0.172%), obtaining labeled data for supervised learning is challenging and often impractical in real-time scenarios.

We treat fraud detection as an **anomaly detection problem**, where the model learns the pattern of normal transactions and identifies unusual (anomalous) transactions as potential fraud.

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## 🎯 Problem Statement

Credit card fraud causes huge financial losses to banks and customers every year. Traditional supervised models require a large number of labeled fraud examples, which are difficult to collect and become outdated quickly.

**Objective:**
Build an end-to-end **unsupervised pipeline** that can effectively detect anomalous transactions without using fraud labels during training.

**Business Impact:**

- Enable real-time fraud monitoring systems for banks
- Reduce financial losses due to fraud
- Improve customer security and trust
- Minimize false positives to avoid unnecessary customer alerts

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## 🛠️ Techniques & Algorithms Used (Syllabus Coverage)

### Unit-wise Coverage:

| Unit | Topics Covered |
|---|---|
| **Unit I** | Foundations, Distance Metrics (Euclidean, Manhattan, Cosine), Data Standardization & Scaling |

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|---|---|
| **Unit II** | k-Means Clustering, k-Means++, Cluster Validation (Elbow Method, Silhouette Score, Davies-Bouldin Index), Inertia |
| **Unit III** | DBSCAN (ε & MinPts tuning, noise points as anomalies), Hierarchical Clustering |
| **Unit IV** | PCA (explained variance, interpretation), t-SNE, UMAP for high-dimensional visualization |
| **Unit V** | **Anomaly Detection**: Isolation Forest, Local Outlier Factor (LOF) |
| **Unit VI** | Evaluation Metrics, Visualization, Interpretability, Ethical Considerations, Dashboard Development |

### Core Algorithms:

- **Clustering**: k-Means, DBSCAN
- **Dimensionality Reduction**: PCA, t-SNE, UMAP
- **Anomaly Detection**: Isolation Forest, Local Outlier Factor (LOF)
- **Visualization**: Matplotlib, Seaborn, Plotly

### Libraries:

- Python, Pandas, NumPy, Scikit-learn
- Matplotlib, Seaborn, Plotly
- umap-learn
- Streamlit (Interactive Dashboard)

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## 📊 Dataset

- **Dataset Name**: Credit Card Fraud Detection
- **Source**: Kaggle (ULB Machine Learning Group)
- **Total Transactions**: 284,807
- **Fraudulent Transactions**: 492 (0.172%)
- **Features**: 30 (Time, Amount + 28 PCA-transformed features: V1 to V28)
- **Download Link**: [https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud](https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud)

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## 🚀 Project Workflow

1. **Data Understanding & Exploratory Data Analysis (EDA)**
2. **Data Preprocessing** (Handling missing values, Scaling using StandardScaler/RobustScaler)
3. **Dimensionality Reduction & Visualization** (PCA, t-SNE, UMAP)

4. **Clustering Analysis** (k-Means, DBSCAN)
5. **Anomaly Detection Modeling**
   - Isolation Forest
   - Local Outlier Factor (LOF)
6. **Model Comparison & Performance Evaluation**
7. **Interactive Streamlit Dashboard**
8. **Business Insights & Ethical Considerations**

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# 📁 Project Structure

```
Credit-Card-Anomaly-Detection/
├── data/
│   └── creditcard.csv
├── notebooks/
│   ├── 01_EDA.ipynb
│   ├── 02_Preprocessing.ipynb
│   ├── 03_Dimensionality_Reduction.ipynb
│   ├── 04_Clustering.ipynb
│   ├── 05_Anomaly_Detection.ipynb
├── src/
├── streamlit_app.py # Interactive Dashboard
├── requirements.txt
├── README.md
└── results/ # Visualizations & Reports
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# 🎨 Key Deliverables

- Comparative study of k-Means vs DBSCAN
- Beautiful 2D/3D visualizations using t-SNE and UMAP
- Parameter sensitivity analysis (ε & MinPts in DBSCAN)
- Isolation Forest vs LOF performance comparison
- Interactive Streamlit Dashboard with multiple tabs:
  - Data Overview
  - Dimensionality Reduction Visuals
  - Anomaly Detection Results
  - Business Insights

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# ⚠️ Ethical Considerations

- Protection of sensitive customer financial data
- Risk of bias in anomaly detection models
- Importance of minimizing false positives (to avoid customer inconvenience)
- Fairness and transparency in automated decision-making

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## 🏆 Learning Outcomes (Aligned with Course Outcomes)

- CO1: Understanding unsupervised learning problem formulation
- CO2: Differentiating clustering paradigms
- CO3: Implementing k-Means, DBSCAN, PCA, t-SNE, UMAP, Isolation Forest, LOF
- CO4: Evaluating models using Silhouette Score, Davies-Bouldin, etc.
- CO5: Building visual analytics dashboard
- CO6: End-to-end project development with insights and ethical discussion

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## 🖥️ How to Run the Project

1. Clone the repository:

   ```
   git clone <repository-link>
   cd Credit-Card-Anomaly-Detection
   ```

2. Install dependencies:Bashpip install -r requirements.txt
3. Run Jupyter Notebooks for step-by-step analysis
4. Launch Streamlit Dashboard:Bashstreamlit run streamlit_app.py